

JKBOSE 2026

CLASS 11TH

STRUCTURE OF ATOM

LECTURE- 07

CHEMISTRY

VISHAL SIR

Bhulchhara
ON
TOP



JKBose Wallah

Topics To Covered

- 1 Quantum Numbers**
- 2 Concept of Orbital**

REVISION.

• Bohr model of atom.

$$\rightarrow E_n = -2.18 \times 10^{-18} \frac{Z^2}{n^2} \text{ J/atom.}$$

$$= -13.6 \frac{Z^2}{n^2} \text{ eV/atom}$$

$$= -1312 \frac{Z^2}{n^2} \text{ kJ/mol.}$$

$$E_1 < E_2 < E_3 < E_4 < E_5 \dots$$

$$\bullet mve = n \frac{h}{2\pi}$$

$$\bullet E_n = a_0 \frac{n^2}{Z}$$

$$a_0 = 52.9 \text{ pm or } 0.529 \text{ \AA}$$

$$\bullet \Delta E = E_f - E_i$$

• Dual nature

$$\lambda = \frac{h}{mv}$$

$$m \uparrow \lambda \downarrow$$

$$v \uparrow \underline{\lambda \downarrow}$$

✓
Heisenberg's
U. P.



Heisenbergs Uncertainty Principle.

It was given by Werner Heisenberg.

It states that "it is impossible to calculate exact position and momentum of a microscopic particle such as e^- ."

Mathematically.

$$\Delta x \cdot \Delta p \geq \frac{h}{4\pi}$$

Δx = uncertainty in position

Δp = Uncertainty in momentum

h = Planck constant

$6.626 \times 10^{-34} \text{ Js}$

OR $6.626 \times 10^{-34} \text{ Kg m}^2 \text{ s}^{-1}$



Heisenberg.

$$\Delta x \cdot \Delta p \geq \frac{h}{4\pi}$$

$$\text{OR } \Delta x \cdot m\Delta v \geq \frac{h}{4\pi}$$

$$\text{OR } \Delta x \cdot \Delta v \geq \frac{h}{4\pi m}$$

$$(p = mv)$$

★

Bright light of high intensity
of high energy

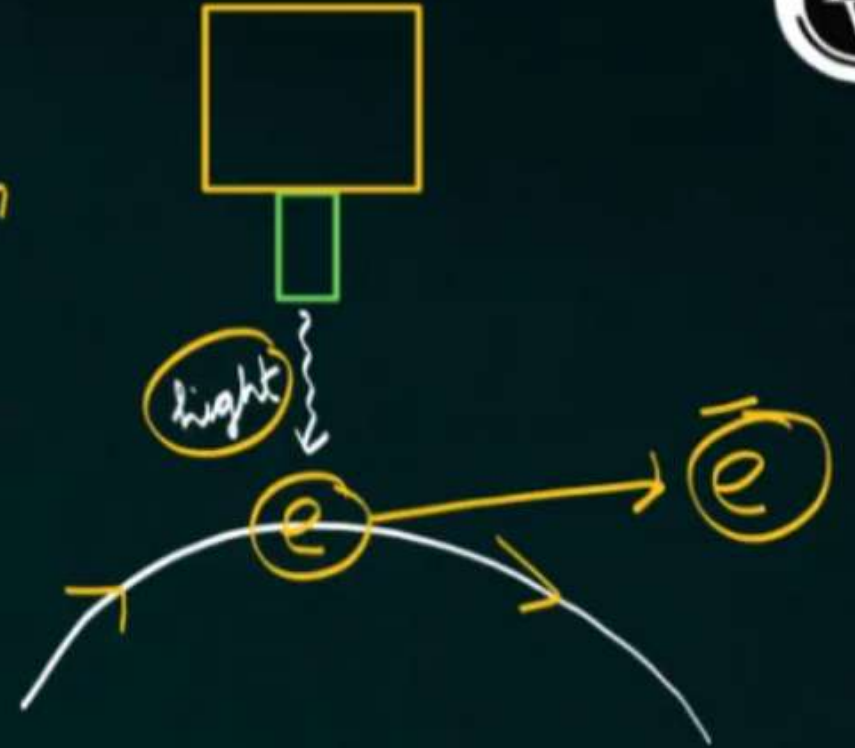
REASON $\Rightarrow$

To determine the position we need light of high intensity (Bright light), but bright light means light of high energy.

and when light of high energy strikes the e^- then its momentum will get uncertain.

And if we use light of less intensity, the position will become uncertain.

or
dim light



QUESTION



What will be the wavelength of a ball of mass 0.1 kg moving with a velocity of 10 ms^{-1} ?

Sol $\lambda = \frac{h}{mv} = \frac{6.626 \times 10^{-34} \text{ Kg m}^2 \text{ s}^{-1}}{m \times v}$

$$\begin{aligned} m &= 0.1 \text{ Kg} \\ v &= 10 \text{ ms}^{-1} \\ \lambda &= ? \end{aligned}$$

QUESTION

$$6.626 \times 10^{-34} \text{ kg m}^2 \text{ s}^{-1}$$

$$1 \text{ nm} = 10^{-9} \text{ m}$$



The uncertainty in the position of an electron is 0.1 nm. What is the uncertainty in its momentum?

$$\Delta x \cdot \Delta p = \frac{h}{4\pi}$$

$$\Delta p = \frac{h}{4\pi \times \Delta x}$$

$$\Delta x = 0.1 \text{ nm} = 0.1 \times 10^{-9} \text{ m} \\ = 10^{-10} \text{ m}$$

$$\Delta p = ?$$

1 $5.27 \times 10^{-25} \text{ kg m/s}$

2 $6.62 \times 10^{-25} \text{ kg m/s}$

3 $1.06 \times 10^{-25} \text{ kg m/s}$

4 $3.29 \times 10^{-25} \text{ kg m/s}$

Quantum Numbers

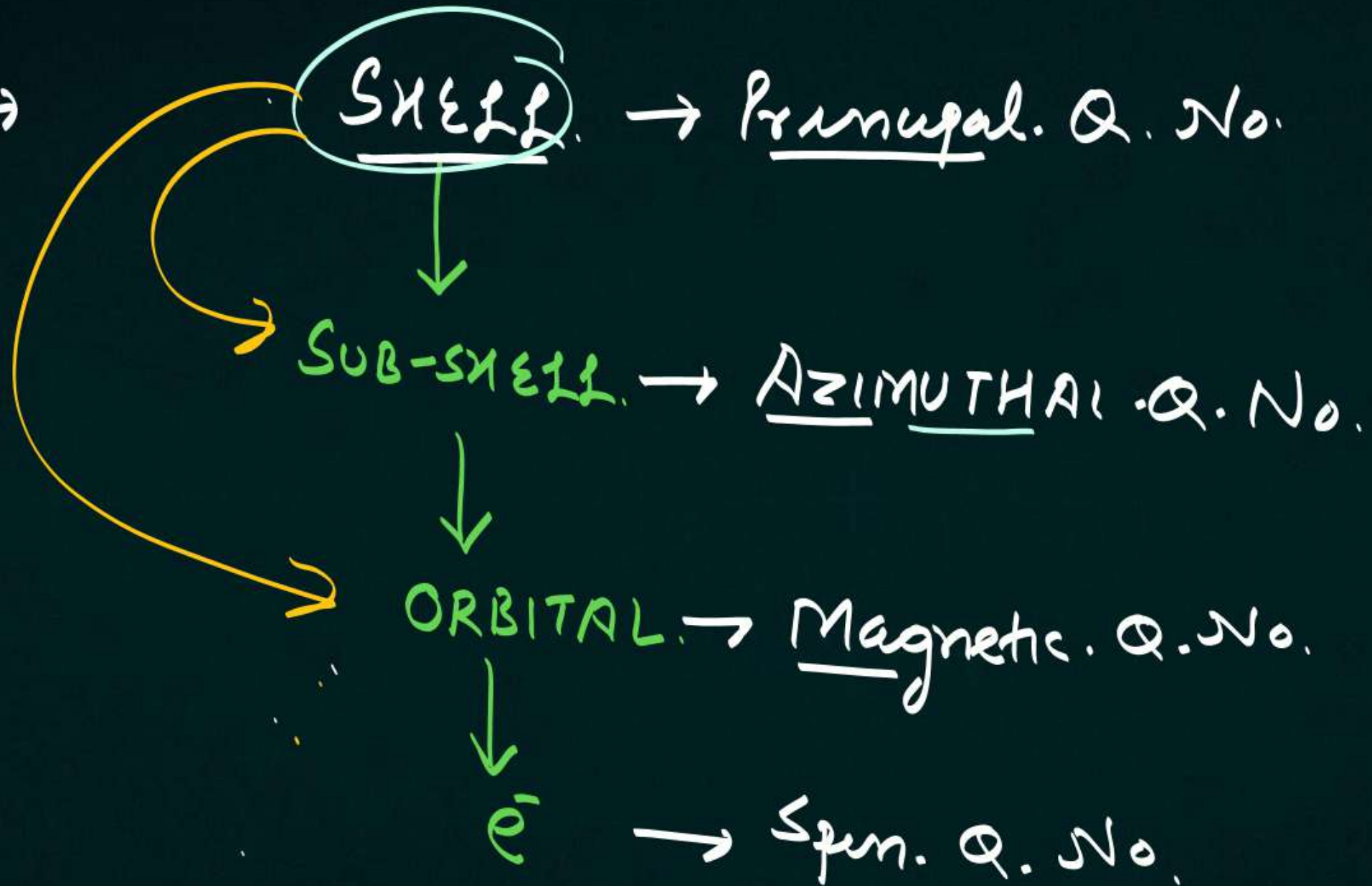
v. Imp



Quantum numbers may be defined as a set of four numbers with the help of which we can get complete information about all the electrons in an atom, i.e., location, energy, the type of orbital occupied, shape and orientation of that orbital, etc.

- (1) Principal quantum number
- (2) Azimuthal quantum number
- (3) Magnetic quantum number
- (4) Spin quantum number

NOTE:→



1) Principal Quantum Number $\rightarrow$

It tells us about in which shell the e is present. It is denoted by 'n'.

$\& n = 1, 2, 3, 4, 5, \dots$

It provides us following information.

- i) It determines radius of shell. $(r_n = a_0 \frac{n^2}{Z})$
- ii) It determines the energy of any shell. $(E_n = -13.6 \frac{Z^2}{n^2} \text{ eV/atom})$
- iii) It determines the no. of e present in shell $= 2n^2$.

e.g. No. of e present in 2nd shell $= 2 \times n^2$
 $= 2 \times 2^2$
 $= 2 \times 4 = 8.$

2) AZIMUTHAL QUANTUM NUMBER OR SUBSIDIARY QUANTUM NUMBER.

It tells us about no. of subshells present in a shell. It is denoted by 'l'.

For any value of 'n', $l = 0$ to $(n-1)$.

e.g. If $n=1$ (1st shell), $l = 0$ (one subshell)

If $n=2$ (2nd shell), $l = 0$ to $(2-1) = 0, 1$ (Two subshells)

$n=3$ (3rd shell), $l = 0$ to $(3-1) = 0, 1, 2$ (Three subshells)

$n=4$ (4th shell), $l = 0$ to $(4-1) = 0, 1, 2, 3$ (Four subshells)
and so on.

i.e.

$l = 0$	1	2	3	4
Name of subshell = s	p	d	f	g

i.e. First shell has s-subshell.

2nd shell has s & p subshell.

3rd shell has s, p & d subshell

4th shell has s, p, d & f subshell etc

- It determines the energy of subshell.
- It determines the angular momentum of e

$$\vec{L} = \sqrt{l(l+1)} \frac{h}{2\pi}$$

Q. Determine the angular momentum of e present in p-subshell? (p-subshell, $l=1$)

Sol.

$$\vec{L} = \sqrt{1(1+1)} \frac{h}{2\pi}$$

$$= \sqrt{2} \frac{h}{2\pi}$$

3) Magnetic Quantum Number

It provides the information about no. of orbitals present in a subshell.

It is denoted by " m_l "

For any value of l , $m_l = (-l \text{ to } +l)$

e.g. If $l=0$ (s-subshell), $m_l = 0$ i.e. one orbital.

If $l=1$ (p-subshell), $m_l = -1, 0, +1$ i.e. 3-orbitals.

$l=2$ (d-subshell), $m_l = -2, -1, 0, +1, +2$ i.e. 5-orbitals

$l=3$, (f-subshell), $m_l = -3, -2, -1, 0, +1, +2, +3$
i.e. seven orbitals.

& so on.

• It determines how many orbitals are present in a subshell = $(2l + 1)$

e.g for s-subshell = $(2\underline{l} + 1) = 2 \times \underline{0} + 1 = \underline{1}$ orbital

p-subshell = $(2 \times \underline{1} + 1) = 3$ orbital.

d-subshell = $(2 \times \underline{2} + 1) = 5$ -orbital & so on.

f-subshell = $(2 \times 3 + 1)$

= $6 + 1$

= 7-orbital.

Note :-

No. of orbitals present in a shell = n^2 .

i.e. for 2nd shell = $2^2 = 4$.

3rd shell = $3^2 = 9$ orbital.

QUESTION



What is lowest value of n that allows g orbitals to exist?

Sol: $n=5$

5th shell, subshells = $s, p, d, f, \underline{g}$

g = subshell.

$l = 0 \quad 1 \quad 2 \quad 3 \quad 4$

$s \quad p \quad d \quad f \quad g$

$n=1 \rightarrow s$

$n=2 \rightarrow s, p$

$n=3 \rightarrow s, p, d$

$n=4 \rightarrow s, p, d, f$

Q.

How many orbitals are present in a fifth shell? ($n=5$)

a) 5

b) 20

c) 25

d) 40

$$n^2 =$$
$$\underline{5^2 = 25 \text{ orbitals}}$$

Q. How many electrons are present in 5th shell?

a) 50

$$2n^2$$

b) 100

$$2 \times 5^2$$

c) 25

$$= 2 \times 25 = \underline{\underline{50}}$$

d) 20.

QUESTION



Using s, p, d, f notations describe the orbital with the following quantum numbers.

Ans

(1) $n = 2, l = \underline{1}$, $\xrightarrow{\text{p}}$ $2p$

(2) $n = 4, l = 0$, $\xrightarrow{\text{s}}$ $4s$

(3) $n = 5, l = 3$, $\xrightarrow{\text{f}}$ $5f$

(4) $n = 3, l = \underline{2}$, $\xrightarrow{\text{d}}$ $3d$

Note:

One orbital has maximum of two e^- .

QUESTION



$$n=3$$

What is the magnetic quantum number m_l for a 3p orbital?

$$\Rightarrow \underline{l=1}, m_l = \underline{\hspace{2cm}} \quad \checkmark$$

$$l, m_l = -l \text{ to } +l$$

$$1, m_l = -1, 0, +1$$

1 0

2 -1, 0, +1

3 -2, -1, 0, +1, +2

4 -3, -2, -1, 0, +1, +2, +3

QUESTION



The maximum number of electrons that can be accommodated in a p-subshell is:

- 1 2
- 2 6
- 3 10
- 4 14

s $\rightarrow$ 1 orbital ✓

p $\rightarrow$ 3 orbitals ✓

d $\rightarrow$ 5 orbitals ✓

f $\rightarrow$ 7 orbitals ✓

$\downarrow$
3 orbitals
 $\downarrow$
no of $e^- = 6$

Q. Calculate the total no. of orbitals in $n = 4$?

a) 16

b) 9

c) 4

d) 12

$$n^2 =$$

$$4^2 = 16$$

Summary



① Aafreena Fayaz ✓

② Mehraan ✓

③ Musaa ✓

REVISE KARNA

Thank
You